QUBO Formulation: Asset Exchange Cycle Selection Problem

Asset Exchange Cycle Selection Problem

Problem Statement. Let $\mathcal{E} = \{0, 1, \ldots, M-1\}$ denote the set of M candidate asset exchanges. Each exchange $i \in \mathcal{E}$ is associated with a positive surplus score s_i. A conflict graph $\mathcal{G} = (\mathcal{E}, \mathcal{C})$ is defined such that an edge $(i, j) \in \mathcal{C}$ exists if and only if exchanges i and j share at least one participant or asset. The objective is to select a subset of exchanges $\mathcal{S} \subseteq \mathcal{E}$ such that no two selected exchanges conflict (i.e., $\mathcal{S}$ is an independent set in $\mathcal{G}$) and the total surplus $\sum_{i \in \mathcal{S}} s_i$ is maximized.

Decision Variables. For each candidate exchange $i \in \{0, 1, \ldots, M-1\}$, introduce a binary decision variable $x_i \in \{0, 1\}$. The variable $x_i = 1$ indicates that exchange i is selected, and $x_i = 0$ indicates it is not.

QUBO Derivation. Step 1 — Write the original objective in general notation and flip sign for minimisation if needed. The original maximization objective is $\max_{\mathbf{x} \in \{0,1\}^M} \sum_{i=0}^{M-1} s_i x_i$. Converting to a minimization problem for the QUBO framework yields:

$$\min_{\mathbf{x} \in \{0,1\}^M} \sum_{i=0}^{M-1} (-s_i) x_i. \tag{1}$$

Step 2 — Write each constraint in its original algebraic form using general indices. The conflict avoidance constraints require that for every conflicting pair $(i, j) \in \mathcal{C}$, we cannot select both exchanges. This is expressed as:

$$x_i + x_j \leq 1, \quad \text{for all } (i, j) \in \mathcal{C}. \tag{2}$$

Step 3 — Convert each constraint to a penalty term. Show the full algebraic expansion. (a) For each constraint $x_i + x_j \leq 1$, we construct a quadratic penalty term using a penalty weight $P > 0$:

$$P(x_i + x_j - 1)^2. \tag{3}$$

(b) Expanding the squared expression in full gives:

$$P(x_i + x_j - 1)^2 = P(x_i^2 + x_j^2 + 1 + 2x_i x_j - 2x_i - 2x_j). \tag{4}$$

Using the idempotent property of binary variables, $x_k^2 = x_k$ for all $k \in \{0, \ldots, M-1\}$, the expression simplifies to:

$$P(x_i + x_j - 1)^2 = P(1 - x_i - x_j + 2x_i x_j). \tag{5}$$

(c) The constant term P is absorbed into the global offset. The linear terms $-Px_i - Px_j$ are absorbed into the linear coefficients of the objective function. The remaining quadratic term $2Px_ix_j$ enforces the conflict constraint. Adjusting the penalty weight scaling to match standard QUBO conventions, the effective penalty contribution for each conflicting pair to the Hamiltonian is Px_ix_j.

Step 4 — Combine objective and all penalty terms into the single QUBO Hamiltonian $H(\mathbf{x})$. Summing the linear objective and all quadratic penalty terms yields:

$$H(\mathbf{x}) = \sum_{i=0}^{M-1} (-s_i)x_i + P \sum_{(i,j)\in\mathcal{C}} x_i x_j. \tag{6}$$

Step 5 — Read off the Q matrix entries and offset c from H in general closed form. The QUBO matrix entries $Q_{i,j}$ and offset c are given by:

$$Q_{i,i} = -s_i, \quad \text{for } i \in \{0, \ldots, M-1\}, \tag{7}$$
$$Q_{i,j} = P, \quad \text{for } (i,j) \in \mathcal{C}, \tag{8}$$
$$Q_{i,j} = 0, \quad \text{for } i \neq j \text{ and } (i,j) \notin \mathcal{C}, \tag{9}$$
$$c = 0.0. \tag{10}$$

Penalty Weight Justification. The penalty weight P must be chosen sufficiently large to dominate any potential gain from violating a constraint. Specifically, we require $P > \max_{i\in\{0,\ldots,M-1\}} s_i$. This condition ensures that the energy cost of selecting any conflicting pair (at least P) strictly exceeds the maximum possible surplus gain from selecting any single exchange. Consequently, any solution containing a conflict will have a higher energy than a corresponding feasible solution with one of the conflicting exchanges removed, guaranteeing that the penalty effectively enforces feasibility.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0,1\}^M$ recovers the optimal exchange selection because (i) for all feasible solutions (where no conflicts exist), all penalty terms vanish, reducing $H(\mathbf{x})$ to $-\sum_{i=0}^{M-1} s_i x_i$, which is minimized exactly when the total surplus is maximized; and (ii) for any infeasible solution containing at least one conflict, the penalty contribution is at least P. By the choice $P > \max_i s_i$, removing one exchange from any conflicting pair reduces the penalty by at least P while reducing the objective by at most $\max_i s_i < P$, strictly lowering the total energy. Therefore, the global minimum of $H(\mathbf{x})$ must occur at a feasible configuration that maximizes the total surplus.

Illustrative Example. Consider a small concrete instance with $M = 3$ candidate exchanges indexed by $i \in \{0, 1, 2\}$, with surplus scores s_0, s_1, s_2. Suppose the conflict graph contains edges $\mathcal{C} = \{(0, 1), (1, 2)\}$, meaning exchange 1 conflicts with both 0 and 2, while 0 and 2 are compatible. Substituting these parameters into the general formulation yields the concrete objective terms $-s_0x_0 - s_1x_1 - s_2x_2$ and the concrete penalty terms $Px_0x_1 + Px_1x_2$. The resulting Hamiltonian is:

$$H(x_0, x_1, x_2) = -s_0x_0 - s_1x_1 - s_2x_2 + Px_0x_1 + Px_1x_2. \tag{11}$$

The corresponding QUBO matrix entries and offset are:

$$Q_{0,0} = -s_0, \qquad Q_{1,1} = -s_1, \qquad Q_{2,2} = -s_2, \tag{12}$$

$$Q_{0,1} = P, \qquad Q_{1,2} = P, \qquad Q_{0,2} = 0, \tag{13}$$

with $c = 0.0$. The optimal selection depends on the relative magnitudes of the surplus scores. If $s_1 > s_0 + s_2$, the optimal configuration is $\mathbf{x}^* = (0, 1, 0)$ with energy $H(\mathbf{x}^*) = -s_1$. If $s_0 + s_2 \geq s_1$, the optimal configuration is $\mathbf{x}^* = (1, 0, 1)$ with energy $H(\mathbf{x}^*) = -(s_0 + s_2)$. In both cases, the solution respects the conflict constraints and maximizes the total surplus.